2.61.

(a) $\bar{E} = \hat{a}_\theta E_\theta = \hat{a}_\theta j\eta \frac{kI_0 l}{4\pi r} e^{-jkr} \sin\theta$.

$= -j\eta \cdot \frac{kI_0 e^{-jkr}}{4\pi r} \hat{a}_\theta l \sin\theta \cdot (-1)$

则 $\bar{l}_e = -\hat{a}_\theta \cdot l \sin\theta$.

(b). $|l_e|_{max} = |-\hat{a}_\theta \cdot l \cdot \sin\theta|_{max}$

When $\theta = \frac{\pi}{2}$, $|l_e|_{max} = l$

(c) $\frac{|l_e|_{max}}{l} = \frac{l}{l} = 1$

2.63 $l = \frac{\lambda}{50}$, $f = 10GHz$.

$\lambda = \frac{c}{f} = \frac{3\times10^8 m/s}{10GHz} = 0.03m. = 3cm.$

因为 $W = \frac{|E|^2}{2\eta_0}$, 故 $E = \sqrt{2\eta_0 \cdot W} = \sqrt{2\times377\Omega\times10^{-3}W/cm^2}$

$= 0.8683 V/cm.$

则 $|V_{oc}|_{max} = |E| \cdot |l_e|_{max}$

$= 0.8683 V/cm \times \frac{3cm}{50}$

$= 0.0521 V$

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2.81 (a). $P_i = \frac{P_{rad}}{4\pi R^2}$ 1 status mile = 1609.3 m.

$= \frac{8W}{4\pi \times (22300 \times 1609.3m)^2} = 4.943 \times 10^{-16} W/m^2$

(b) $f = 2GHz$. $\lambda = \frac{c}{f} = \frac{3\times10^8 m/s}{2\times10^9 Hz} = 0.15m$.

$D_0 = 60dB = 10^6$.

$A_{em} = \frac{\lambda^2}{4\pi} \cdot D_0 = \frac{(0.15m)^2}{4\pi} \times 10^6 = 1790.49m^2$

$P_r = A_{em} \cdot P_i = 1790.49m^2 \times 4.943 \times 10^{-16} W/m^2$

$= 8.85 \times 10^{-13} W$.

2.95. $P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$, $\sigma = \frac{P_r (4\pi)^3 R^4}{P_t G^2 \lambda^2}$, $\lambda = \frac{c}{f} = 1m$.

$\sigma = \frac{0.1425\times10^{-3} \times (4\pi)^3 \times (500)^4}{10000 \times 75 \times 75 \times 1^2} = 3141.96m^2$

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